

AI/Humanoid Workbench

Open Source Concept — Prior Art Disclosure

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This document is released into the public domain as a Defensive Publication. Its purpose is to establish prior art and prevent any party from claiming patent ownership over the general concept of specialized workstation systems designed for artificial intelligence robotic systems.

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1. Executive Summary

The integration of artificial intelligence robotic systems into real working environments faces a critical challenge: how can AI-driven robots use standard tools — designed for human hands — in practical, productive ways?

This document defines the concept of an AI Workbench, its scope, the key technical challenges it addresses, the potential solution space, relevant market context, and the reasoning behind its release as public domain prior art. It also addresses the critical legal question of what constitutes an AI system in this context — a question that remains unresolved across major regulatory jurisdictions.

An AI Workbench is a specialized workstation system that provides an ergonomic and productive environment optimized for the physical characteristics, kinematic capabilities, and limitations of artificial intelligence robotic systems — enabling them to work effectively with standard or existing tools, infrastructure, and workflows.

The Humanoid Workbench is a specific implementation of an AI Workbench optimized for humanoid robot form factors.

Key Points

- The concept applies to all AI robotic systems, with humanoid robots as the primary use case
- The environment is adapted to the AI robotic system — not the other way around
- Addresses both hardware and workflow challenges in AI-driven tool use
- Released as public domain to prevent patent monopolization of the general concept
- Applicable across manufacturing, maintenance, construction, and service industries
- The legal definition of "AI system" remains contested across jurisdictions — this document establishes prior art regardless of how that definition evolves

2. Market Context & Relevance

The AI robotics industry is entering a period of rapid commercialization. Companies such as Tesla (Optimus), Figure AI, Agility Robotics, Boston Dynamics, and Apptронik are actively deploying or developing AI robotic platforms for industrial use. A central challenge in nearly every deployment scenario is tool use.

Why Tool Use Is the Critical Bottleneck

Human workers have spent millennia co-evolving with their tools. Every screwdriver, drill, wrench, and power tool on the market today is designed around the human hand — its size, grip strength, dexterity, and proprioception. AI robotic systems face an immediate disadvantage:

- Hand/gripper geometry differs — AI robot end-effectors have different dimensions and compliance characteristics than human hands
- Force modulation is imprecise — applying exactly the right torque or pressure remains difficult
- Tactile feedback is limited — most AI robots cannot yet feel slippage or resistance the way humans do
- Tool retrieval is unguided — picking up a loose screwdriver from a surface remains surprisingly complex
- Battery and accessory changes require fine motor skills that exceed current AI robot dexterity

Deployment Scale

Analysts project that the AI robotics market will grow from approximately \$1.5B in 2024 to over \$38B by 2035. Every deployed unit in an industrial setting will require some form of tool interface. The AI Workbench concept addresses this at a systemic level — as infrastructure, not just as a per-robot adaptation.

Industry Sector	Primary Use Cases
Manufacturing	Assembly, fastening, quality inspection, maintenance
Automotive	Component installation, torque-critical fastening, paint prep
Construction	Drilling, cutting, measurement, material handling
Logistics & Warehousing	Packaging, sorting, labeling, pallet building
Maintenance & Repair	Infrastructure inspection, repair, tool-based servicing
Healthcare & Laboratories	Instrument handling, sample processing, sterile assembly

3. Legal Definition Framework — Key Unresolved Questions

Across the European Union, United States, and Japan, three different AI regulatory frameworks exist. None adequately address the specific case of AI systems designed to enable tool use in workbench environments.

3.1 How Each Jurisdiction Defines "Artificial Intelligence"

European Union — AI Act (2024)

A machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment, and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments.

Key elements: machine-based, varying autonomy levels, adaptiveness, inference capability, output generation that influences environments.

Liability framework: High-risk AI systems require third-party conformity assessment, post-deployment monitoring, and mandatory human oversight.

OECD Definition (Adopted by 47 Countries)

A machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments. Different AI systems vary in their levels of autonomy and adaptiveness after deployment.

Key distinction from EU: Does not explicitly require "designed to operate with varying autonomy" — focuses on capability rather than intent.

United States — Current Status (2026)

The United States has no single comprehensive federal AI definition. Multiple frameworks exist:

- White House National Policy Framework (2026) references trustworthy AI but provides no binding legal definition
- Individual states (California, Colorado, others) have adopted variations on the EU or OECD definitions
- Federal agencies apply sector-specific definitions (healthcare, finance, autonomous vehicles)
- U.S. courts currently treat AI-caused harm under existing product liability, negligence, and agency law

Liability determination: Case-by-case through existing tort and contract law, with no unified classification standard.

Japan — AI Promotion Act (2025)

Technologies that replicate human intellectual capabilities like cognition, inference, and judgment through artificial means, as well as the systems that use them.

Key distinction: Emphasizes replication of human cognition rather than autonomy or adaptiveness.

Liability framework: Operators remain liable for outcomes regardless of AI involvement (strict operator liability model).

3.2 Critical Gaps in Current Definitions

Gap 1: Local vs. Cloud-Based AI Decision-Making

None of the major definitions specify whether an AI system must be locally executed (on-robot processor), cloud-connected (remote server), or hybrid. The location of the "intelligence" — local

processor vs. remote server — is not addressed in any existing framework. Current regulatory treatment: Silent across EU AI Act, OECD, U.S. case law, and Japan.

Gap 2: Autonomy Levels and Control Thresholds

The EU and OECD definitions reference "varying levels of autonomy" but do not define what constitutes "autonomy" or at what point a system transitions from "tool" to "autonomous agent." No jurisdiction provides a legal threshold.

Gap 3: Post-Deployment Adaptiveness

The EU and OECD definitions require that AI systems "may exhibit adaptiveness after deployment." But "adaptiveness" is not legally defined. No jurisdiction provides a threshold distinguishing learning from data collection.

Gap 4: Inference in Tool-Use Contexts

All definitions reference systems that "infer" outputs, but do not define what constitutes "inference" in tool-use contexts. Selecting a tool, adjusting grip force, or predicting task sequences may or may not qualify. No jurisdiction provides guidance.

Gap 5: Physical Influence and Shared Human-AI Control

All definitions reference systems that "influence physical environments," but do not address shared control scenarios. Liability in hybrid human-AI operations is undefined across all three jurisdictions.

3.3 The Five Critical Unresolved Questions for Courts and Regulators

Question 1: Control Location and Legal Status. Does the legal classification of an AI system change based on whether decision-making occurs locally versus remotely (cloud server)? At what point does a robot receiving real-time cloud AI commands become "autonomous" rather than "remote-controlled"?

Question 2: Liability in Shared Human-AI Control. When a humanoid robot and a human operator share control of a task — human holding a workpiece while robot operates a power tool — who bears liability for errors? Does the workbench design itself carry liability obligations?

Question 3: Autonomy Threshold. At what level of autonomy does a system transition from "tool under human control" to "autonomous agent"? Does this transition trigger different legal obligations for manufacturers, operators, and workbench designers?

Question 4: Post-Deployment Adaptiveness and Learning. Does a robot that learns from user feedback during deployment constitute an "adaptive AI system" under EU or OECD definitions? Or is this simply data logging and parameter adjustment?

Question 5: Workbench Design and Liability. If an AI Workbench contains AI-based guidance systems (vision alignment, force feedback interpretation, task sequencing), is the workbench designer liable for AI system errors?

4. Scope Definition

An AI Workbench is a specialized workstation system optimized for artificial intelligence robotic systems of any form — including but not limited to humanoid robots, robotic arms with AI decision-making, mobile manipulators, and other AI-driven robotic platforms.

4.1 What Falls Within Scope

- Physical workbench surfaces with integrated features (guidance systems, fixturing)
- Tool storage, staging, and retrieval systems optimized for AI robot kinematics
- Docking stations and mechanical interfaces for tool exchange
- Force and torque absorption and guidance systems
- Vision and sensor-guided alignment systems for tool-workpiece interaction
- Software and control systems that coordinate AI robot-tool interaction
- Hybrid human-AI robot shared workstations
- Tool staging systems designed around AI robot reach envelopes
- Fastener presentation systems (feeders, orientation fixtures, magnetic guides)

4.2 Scope Boundaries

The concept does NOT apply to:

- General-purpose industrial robot workcells designed for traditional industrial arms
 - Standard human workbenches not adapted or designed with AI robotic use in mind
 - Systems that require tools to be fundamentally redesigned for robotic use
 - Workbenches designed solely for manual human work without AI robotic integration
 - Traditional robot tool changers not designed for standard human tools
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5. Core Principles

The AI Workbench concept starts from a simple insight: the most practical and scalable way to make AI robotic systems productive is to design the environment around them — not to wait for perfect dexterity, and not to redesign every tool from scratch.

5.1 Adapt the environment, not the tool. Standard tools should remain unmodified and widely available.

5.2 Bridge the gap. Compensate for the differences between human hands and AI robot end-effectors through intelligent workbench design.

5.3 Multiple solutions welcome. There is no single correct way. Adapters, docking stations, mechanical aids, automated systems, or completely new approaches are all valid.

5.4 Keep it practical. The workbench should make AI robotic systems useful in real workshops, small businesses, and field environments — not just laboratories.

5.5 Open for everyone. The core concept must stay open so small workshops, craftspeople, startups, and individuals are not locked out by patents or licensing barriers.

6. Key Technical Challenges

Challenge	Root Cause	Impact
Trigger & Switch Operation	End-effector geometry mismatch, limited finger/gripper force	Cannot activate power tools reliably
Battery Exchange	Fine motor tolerance, visual alignment	Cordless tools become unusable mid-task
Accessory Changes	Chuck/socket release forces, small part grip	Cannot change drill bits or sockets
Fastener Handling	Picking small parts, maintaining orientation	Fasteners dropped, misaligned, cross-threaded
Workpiece Fixturing	Dual-hand coordination, compliance mismatch	Workpiece moves during operation
Force Control	Limited torque sensing, joint compliance	Under/over-torquing, tool damage
Tool-Workpiece Alignment	Proprioception limits, visual system latency	Missed holes, surface damage
Tool Switching	Grasp release, re-grasp, storage coordination	Slow cycle times, task interruption
Tool Retrieval	Unguided pickup from flat surfaces	Reliably picking up tools without guidance
Two-Handed Operations	Coordination between multiple arms/manipulators	Cannot perform dual-arm task sequences

7. Solution Space

This document does not define specific solutions. It defines categories of solution approaches that fall within scope. Any implementation addressing the challenges in Section 6 — or through novel approaches not listed here — constitutes an AI Workbench.

7.1 Mechanical Interface Solutions

- Tool docking stations that present tools in fixed, pre-aligned orientation
- Trigger and switch adapter mechanisms that translate AI robot grip/force into tool actuation
- Battery exchange fixtures that guide insertion and removal within AI robot tolerances
- Chuck and socket change systems with AI robot-compatible release mechanisms
- Fastener presentation systems (screw feeders, orientation fixtures, magnetic guides)

7.2 Workbench Surface Solutions

- Integrated fixturing and clamping systems operable by AI robot end-effectors
- Force and torque absorption surfaces for drilling and fastening operations
- Modular tool storage integrated into the work surface
- Guided approach channels for tool pickup and placement
- Compliance-matched work surfaces that compensate for AI robot joint stiffness

7.3 Sensing & Software Solutions

- Vision-guided tool alignment systems (camera-based approach correction)
- Force-torque feedback integration for controlled fastening operations
- Task-level software coordination for tool selection and sequencing
- Digital twin integration for pre-planned tool interaction trajectories
- Real-time adjustment systems based on sensor feedback from the workbench

7.4 Hybrid & Novel Solutions

- Human-AI robot collaborative fixturing (human holds workpiece, AI robot operates tool)
- Modular adapter ecosystems compatible across tool brands and AI robot platforms
- Solutions combining multiple categories above
- Approaches not yet conceived at the time of this publication

This list is illustrative, not exhaustive. The purpose of defining solution categories is to establish the breadth of the prior art claim — not to limit implementations to only these approaches.

8. Prior Art & Related Work

Related Concept	Distinction from AI Workbench
Standard industrial robot workcells	Designed for traditional industrial arms, not AI robotic systems or standard tools
Tool changers for industrial robots	Proprietary end-effector swaps, not standard human tools
Human workbenches	Optimized for human ergonomics, not AI robotic reach or grip
Cobots (collaborative robots)	Focus on safety in shared spaces, not tool interface optimization
Robot-specific tool designs	Redesigns the tool, not the workstation environment
Assembly jigs and fixtures	Workpiece-oriented, not tool-interface-oriented
Autonomous vehicle charging stations	Vehicle-specific infrastructure, not tool-use infrastructure

To the best of the author's knowledge, no prior published concept, patent, or system specifically addresses the problem of workstation design optimized for AI robotic tool use as a general category, as defined in this document.

9. Application Scenarios

9.1 Industrial Assembly

An AI robotic system performs multi-step assembly on a production line. The AI Workbench provides pre-staged tools in AI robot-accessible positions, a fastener feed system, and fixturing for the

workpiece — enabling the robot to complete the assembly cycle without human assistance or custom per-robot programming.

9.2 Maintenance & Repair

An AI robotic system performs scheduled maintenance on industrial equipment. The workbench provides a tool staging area near the work site, with battery swap capability for cordless tools and guided retrieval for socket sets and torque wrenches.

9.3 Construction Site

AI robotic systems work alongside human crews on a construction site. Shared workbenches positioned throughout the site allow both human and AI robot workers to access and return tools — with AI robot-specific guidance features active when a robot is operating and inactive (or retracted) when a human approaches.

9.4 Small Workshop / SME

A small manufacturing business deploys one or more AI robotic systems for overnight or extended-shift work. A compact AI Workbench occupies one corner of the workshop, providing the robot with everything it needs to operate standard tools through a full shift without human supervision.

9.5 Field Maintenance and Remote Operations

An AI robotic system is deployed to a remote site for equipment maintenance or repair. A portable AI Workbench containing standard tools and guidance systems allows the robot to perform complex tasks without human on-site presence, coordinated remotely via cloud-based AI decision-making.

10. The Patent Risk — Why This Document Exists

The AI Workbench concept is being published as prior art for a specific reason: the risk that one or more large corporations could file broad patents on the general concept of adapting work environments for AI robotic tool use — and use those patents to lock everyone else out.

This Is Not Hypothetical. Broad software and system patents have repeatedly been used to block entire industries. Amazon patented 1-Click purchasing. Rambus patented memory interface standards while sitting on standards committees. Patent thickets in robotics already create significant barriers for smaller players. This is a well-documented pattern.

What Could Be Patented — and What That Would Mean

Without this prior art on record, a corporation could file broad patents such as:

- "A workstation system optimized for AI robotic tool use"
- "A docking mechanism for presenting hand tools to an AI robot end-effector"
- "A method for adapting a work environment for AI robotic integration"
- "A tool staging system designed around AI robotic reach envelopes"
- "A workbench control system for coordinating AI robot task execution"

If granted, patents like these would give a single company the legal right to demand licensing fees from anyone who uses an AI robotic system with standard tools in a workbench setting. That means:

- Small workshops and craftspeople could be forced to pay royalties just to deploy an AI robot
- Manufacturers integrating AI robots into existing production lines could face legal action
- Startups building workbench solutions for AI robots could be sued out of existence
- Individuals and small businesses would be priced out — only large corporations could afford the license
- Existing workflows and infrastructure could become inaccessible to AI robots without paying a gatekeeper

The core danger is not just financial. It is structural: a patent on this concept would hand control over how AI robots integrate into real working environments to a single private entity. Small workshops, craftspeople, repair shops, and small manufacturers — the people who would benefit most from affordable AI robot assistance — would be locked out entirely.

How This Document Prevents That

- This document is timestamped, authored, and publicly accessible — it meets the standard for prior art
- It covers the general concept broadly and intentionally, to make narrow workarounds harder
- It explicitly names the problem space so that trivial variations cannot be claimed as novel
- It is archived and version-controlled so it cannot be disputed or removed

This does not prevent all patents in this space. Genuinely novel technical solutions — a specific mechanism, a particular software system, a unique adapter design — can still be protected. What cannot be protected is the general concept itself.

11. Public Domain Declaration

This document constitutes a Defensive Publication — a formal mechanism for placing an inventive concept into the public domain in order to establish prior art and prevent future patent claims on the same general idea.

Legal Basis

Under patent law in most jurisdictions, a patent cannot be granted for an invention that was publicly disclosed before the patent application was filed. By publishing this document with clear authorship, date, and public accessibility, the concept of the AI Workbench / Humanoid Workbench enters the prior art record as of June 2026.

Any patent application filed after this date that claims the general concept of a workstation system specifically designed to enable AI robotic systems to use standard tools should be challengeable on the basis of this prior art disclosure.

This document is released into the public domain. Anyone is free to implement, commercialize, build upon, or otherwise use the concepts described herein. No attribution is required, though it is appreciated. No permission is needed.

Core Intent

The explicit goal of this prior art disclosure is to keep the entire solution space open. Nobody should be able to patent the general concept of helping AI robotic systems use standard tools or work within existing infrastructure — and thereby block others from building solutions in this space.

Anyone — individuals, startups, companies, researchers — must remain free to develop, build, sell, and deploy any system that helps AI robotic systems use standard tools or integrate into existing working environments. No permission, license, or royalty should ever be required for the general concept.

What This Does NOT Restrict

- Specific highly novel implementations with unique technical inventions may still be independently patentable — but the general concept cannot be
- Software, specific mechanical designs, and particular system architectures with genuine inventive step remain protectable
- Trademarks, branding, and trade secrets are unaffected by this disclosure
- Commercial development of AI Workbench and Humanoid Workbench products is explicitly encouraged — this disclosure exists to enable that, not restrict it

12. Document Versioning & Updates

Version	Notes
1.0 — June 2026	Initial public release. Humanoid-specific definition, challenges, solution space, and prior art declaration.
2.0 — June 2026	Expanded scope to cover all AI robotic systems. Added Section 3: Legal Definition Framework comparing US, EU, and Japan AI definitions. Five critical unresolved questions for courts and regulators.
Future versions	May include additional application scenarios, updated market context, case law analysis, or expanded solution categories as the field develops.

The latest version of this document is maintained at humanoidworkbench.com. All versions are archived and timestamped for prior art purposes.

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